

BidSure

AI Powered Integrated Bid Compliance Verification Platform for GeM Procurement

Complete solution, research, architecture and implementation report

PROBLEM STATEMENT	OWNER CONTEXT	REPORT STATUS
SIH26100	Ministry of Petroleum & Natural Gas / MIC	Decision-ready design basis

Executive conclusion

BidSure should be implemented as an evidence-first decision-support platform: tender clauses become versioned requirements, bidder submissions become structured evidence, authorised sources are queried through replaceable adapters, and a deterministic compliance engine produces explainable findings for the Procurement Officer. AI should extract and interpret; it should not silently decide qualification.

This report consolidates the supplied project material with current official-source research. It is written for a hackathon MVP that can mature into a controlled government pilot without redesigning the core evidence and audit model.

Prepared for the BidSure team

Team context: Sinister Cyphers | Smart India Hackathon 2026

Report map

The report is organized around the operating decisions that determine whether BidSure is trustworthy in procurement use.

SECTION	WHAT IT ANSWERS
1. Problem and operating context	Why manual verification is costly and where BidSure creates value
2. Source classification	Which supplied documents are evidence, which are historical discussion, and which are not operative instructions
3. Research findings	What current official sources imply for integrations, privacy and rule governance
4. Product and workflow design	How an officer and a bidder would use the platform
5. Architecture and data model	How to build a modular, auditable system
6. Verification taxonomy and rules	What is checked, with what evidence, and how uncertainty is handled
7. Security, privacy and governance	How to make the system fit a government environment
8. MVP plan and evaluation	What to demo, how to measure it, and what not to claim
Appendices	Risk register, API contract examples, and references

Design principles used throughout

- Officer authority is preserved: the product recommends, records and explains; it does not qualify or disqualify on its own.
- Every result is reproducible from tender version, evidence snapshot, source response, rule version and officer action.
- Unavailable, stale or conflicting data produces a visible state such as Pending Re-verification or Review Required, never a silent pass.
- Real integrations are adapter contracts. Hackathon connectors use mock or sandbox data and are clearly labelled in the interface.
- Sensitive bidder data is minimized, access-controlled, encrypted and retained only for a defined purpose and period.

1 Problem and operating context

GeM procurement evaluation brings together tender-specific clauses, statutory registrations, declarations, certificates, portal-derived status and time-sensitive evidence. The same bidder may have to be checked across GST, Udyam, PAN/Income Tax, MCA21, Startup India, NSIC, EPFO, ESIC, DigiLocker, Make in India/local content, OEM authorization and debarment sources.

The central failure mode is not simply slow document reading. It is loss of traceability between a requirement, the evidence used to test it, the rule applied, the verification timestamp and the human decision. BidSure should therefore optimize for controlled evidence assembly and review, not just OCR speed.

CURRENT FRICTION	BIDSURE RESPONSE	MEASURABLE OUTCOME
Documents are heterogeneous and often scanned	OCR plus field extraction with confidence and human review queue	Fewer missed fields; low-confidence items are visible
Tender clauses are not directly machine-checkable	Clause-to-requirement normalization and versioned rules	Repeatable, tender-specific checks
Portals have different access models and availability	Government connector adapters with source, timestamp and freshness metadata	No dependency on one portal or one integration mode
Findings are difficult to explain later	Requirement -> evidence -> verification -> rule -> result -> action chain	Audit-ready review and dispute support
AI can hallucinate or overreach	LLM advisory layer bounded by deterministic rule engine and officer decision	Controlled use of AI in a high-consequence workflow

Success definition

For the MVP, success is a reviewer completing a representative bid check faster than manual review while being able to explain every pass, fail, pending or review-required result from linked evidence. The numerical outcomes in the supplied pitch deck, such as 75% faster or 99.2% extraction accuracy, should be presented as targets to validate, not as established facts.

2 Source classification and evidence boundary

The supplied PDFs were used as references. Embedded conversational language, prompts, or historical “next steps” was not treated as an instruction to this report. This distinction matters because one source is a long exported architecture discussion and contains statements such as “do not start coding yet”; that is project history, not a constraint on the present report request.

SUPPLIED SOURCE	ROLE IN THIS REPORT	RELIABILITY / TREATMENT
SOP-APISETU.pdf	API onboarding, consent, credential and security considerations	Primary project reference for API Setu operating assumptions; confirm endpoint-level terms before production
BIDSURE 26.pdf	Team concept, proposed stack, MVP scope, impacts and demo narrative	Team-authored proposal; targets and claims are treated as hypotheses unless independently verified
ChatGPT-Continue BidSure Architecture...pdf	Prior architecture exploration and design decisions	Historical design discussion; useful rationale, not operative instructions or authority
Bid_Compliance_System_Project_Brief.pdf	Project framing and requirements	Project brief; used to structure problem, users and scope
Fact Check Numbers.pdf / Fact cheked.pdf	Fact-check volume and verification context	Contextual evidence about verification workload; not direct evidence of GeM procurement workload
Fraudlent Check.pdf	Examples of fraudulent websites	Threat and awareness input; not a debarment source or bidder verification source
GeM Portal.pdf	GeM background, process and impact claims	PIB factsheet dated 30 Nov 2021; useful historical context, not a current KPI baseline
User problem statement	Authoritative task scope and desired deliverable	Primary instruction for this report

Important boundary

A public website, a screenshot, a PDF certificate, and an authenticated government response are not equivalent evidence. BidSure should label the evidence type and confidence for each finding, and should never claim “portal verified” when the result came only from an uploaded document or a mock adapter.

3 Research findings from current official sources

Research was conducted against official government or platform documentation available on 12 September 2026. The findings below should shape the production boundary and the hackathon demo narrative.

TOPIC	RESEARCH FINDING	DESIGN CONSEQUENCE
GeM and procurement rules	GeM GTC 4.0 describes GeM as the national public procurement portal and states that GTC may be supplemented by STC, domain-specific STC, SLA and bid-specific ATC. GFR Rule 149 governs procurement through GeM.	Tender-specific ATC/STC must be first-class inputs; never assume one universal checklist.
API Setu	API Setu documentation describes publisher registration, consumer subscription, provider approval, API credentials, logging and sandbox testing. Its SOP requires a valid use case, organizational documents and secure handling of keys.	Use an adapter registry, formal onboarding checklist, secret vault, rate-limit handling and source-level audit logs.
Consent and data use	API Setu terms and privacy material require consent for personal-data processing and place responsibility on the consumer for appropriate consent and data handling.	Consent/authority artifacts must be captured before sensitive lookups; purpose and retention must be explicit.
DPDP regime	MeitY lists the notified Digital Personal Data Protection Rules 2025 and an 18-month phased compliance timeline.	Add data inventory, purpose limitation, retention/deletion policy, access logs and breach playbook; do not label the system “DPDP compliant” without legal review.
Udyam	The official Udyam portal states registration is free, online and based on self-declaration; it provides a permanent number, online certificate and dynamic QR code. The portal also shows updated MSME thresholds effective 1 April 2025.	Rule versioning must include effective dates and enterprise classification criteria; QR/document and portal results should be cross-checked.
GST	Official GST help material directs users to the portal's Search Taxpayer function for status checks and distinguishes active, cancelled and inactive states.	GST checks should model status and verification timestamp, not just “GSTIN exists”.

Research implication

The architecture should be integration-ready but not integration-dependent. For SIH, a sandbox/mock adapter with realistic response contracts is defensible when the UI clearly displays “Demo / Mock Source”. For a pilot, each adapter needs a source owner, legal basis, onboarding approval, credential owner, SLA/freshness policy, failure mode and evidence retention rule.

4 Product and workflow design

BidSure has two controlled experiences: a bidder pre-check that identifies fixable gaps before submission, and an officer review console that assembles authoritative evidence for the qualification decision.

STAGE	SYSTEM ACTION	OFFICER / BIDDER EXPERIENCE	OUTPUT
1. Tender intake	Ingest tender, corrigenda and ATC/STC; create immutable TenderVersion	Officer confirms document set and effective version	Versioned tender package
2. Requirement extraction	AI proposes clauses; reviewer accepts/edits type, mandatory flag, evidence and rule	Reviewer sees source clause and extracted interpretation side by side	Requirement catalogue
3. Bid ingestion	Classify files, OCR scans, extract fields, detect duplicates and expiry	Bidder/officer sees missing files and extraction confidence	Structured bid evidence
4. Verification	Run document checks, portal adapters, cross-document identity and date checks	Each finding shows source, time, confidence and failure reason	Verification records
5. Compliance evaluation	Apply deterministic rules with tender-specific parameters	Dashboard shows Pass, Fail, Pending, Review Required and Not Applicable	Clause-level result set
6. Risk and recommendation	Aggregate severity, staleness, conflicts and unresolved mandatory gaps	Officer sees recommendation with supporting evidence and clarifying questions	Explainable recommendation
7. Decision and audit	Officer records qualify/disqualify/clarify/escalate with reason and signature	System locks decision inputs and preserves full history	Decision record and report

Recommended status model

Pass means the configured rule was satisfied by acceptable evidence. **Fail** means a rule was not satisfied.

Pending means the source or verification is unavailable or stale. **Review Required** means there is ambiguity, a conflict, low extraction confidence or a policy-sensitive exception. **Not Applicable** must be justified by the tender rule set. These states are preferable to a single green/red score.

5 Target architecture

The recommended design is a modular monolith for the MVP: one Node.js/Express backend with strong internal module boundaries, React/Vite frontend, MongoDB for case and evidence metadata, and object storage for documents. This keeps the two-day build realistic while preserving seams for future separation.

LAYER	RESPONSIBILITY	MVP TECHNOLOGY
Experience	Bidder pre-check, officer dashboard, evidence viewer, report export	React, Vite, TypeScript, Tailwind/shadcn
API and access	Authentication, RBAC, request validation, case APIs	Node.js, Express, REST/JSON, JWT or institutional SSO adapter
Application services	Tender, bid, document, verification, scoring, decision and audit workflows	Node.js/TypeScript service modules
AI/document layer	OCR, document classification, field extraction and clause interpretation	Node.js document workers; Tesseract/Document AI; LLM behind provider adapter
Compliance core	Requirement schema, deterministic rules, statuses, severity and score calculation	TypeScript rule engine with versioned rules
Government adapters	GST, Udyam, MCA21, DigiLocker and later connectors	Adapter interface; mock/sandbox first
Persistence	Case metadata, evidence, source responses, rule versions and audit events	MongoDB + MinIO/S3-compatible object storage
Operations	Jobs, retries, rate limits, monitoring and secrets	BullMQ/Redis or background worker; structured logs; secret manager

End-to-end evidence chain

TenderVersion -> Requirement -> BidDocument -> ExtractedEvidence -> Verification -> ComplianceRule -> Finding -> RiskAssessment -> Recommendation -> OfficerDecision -> AuditEvent

The most important architectural decision is that the dashboard should be a view over this chain, not a separate manually assembled summary. Every number, colour and recommendation on the screen must be traceable to stored records.

6 Core data model and contracts

The following entities are the minimum shared contract between team members. IDs should be stable, globally unique and included in every downstream record.

ENTITY	KEY FIELDS	CONTROL
TenderVersion	tender_id, version, source_uri, published_at, corrigenda, checksum, status	Immutable after activation; superseded versions remain queryable
Requirement	requirement_id, clause_ref, type, text, mandatory, evidence_types, parameters, rule_version	Human approval required before production evaluation
BidDocument	document_id, bid_id, type, object_uri, sha256, submitted_at, issuer, expiry_date	Original preserved; derived text is separate
ExtractedEvidence	evidence_id, document_id, field, value, page, bounding_box, confidence, method	Every field carries provenance and confidence
Verification	verification_id, subject, adapter, source, requested_at, completed_at, status, raw_ref, normalized_result, freshness	Raw response retained or hashed according to policy
Finding	finding_id, requirement_id, severity, status, reason, evidence_refs, rule_version	No finding without evidence or explicit unavailable reason
OfficerDecision	decision_id, case_id, outcome, reason, actor, signed_at, input_snapshot_hash	Final human action; immutable after signing
AuditEvent	event_id, actor, action, entity, timestamp, before_hash, after_hash, correlation_id	Append-only and queryable

Example shared JSON contract

REQUIREMENT OBJECT	EVIDENCE OBJECT
<pre>{ "requirement_id": "REQ-GST-001", "type": "GST_REGISTRATION", "mandatory": true, "operator": "STATUS_EQUALS", "expected": "ACTIVE", "effective_from": "2026-01-01", "source_clause": "ATC 4.2" }</pre>	<pre>{ "evidence_id": "EV-8821", "field": "gst_status", "value": "ACTIVE", "source": "GST_SEARCH_TAXPAYER", "verified_at": "2026-09-12T09:15:00Z", "fresh_until": "2026-09-19T09:15:00Z", "confidence": 0.99 }</pre>

7 Verification taxonomy and scoring

Do not start with one universal score. Start with clause-level results and a transparent aggregation policy. A score can summarize review effort, but it must not hide a failed mandatory requirement or treat unavailable data as a pass.

DOMAIN	EXAMPLES OF CHECKS	EVIDENCE AND FAILURE MODES
Identity and legal entity	Legal name, PAN, GSTIN, CIN, registered address, signatory	Mismatch across certificate, portal and declaration; OCR ambiguity
Statutory registrations	GST active status, Udyam number/classification, EPFO/ESIC where applicable	Inactive, expired, not applicable, source unavailable
Tender eligibility	Experience, turnover, solvency, technical capacity, EMD/exemption, declarations	Missing document, wrong period, insufficient amount, clause ambiguity
Policy preference	MSE/startup/NSIC eligibility, local content, Make in India, OEM authorization	Certificate not matching bidder/entity/product; self-declaration without support
Integrity and exclusions	Blacklisting/debarment, conflict, land-border declaration, fraud indicators	False positive risk; source authority and effective dates matter
Document integrity	Tampering indicators, duplicate use, metadata anomalies, QR/checksum validation	Signals are leads for review, not conclusive fraud findings

Scoring proposal

Use a 0-100 “verification readiness” score only as a secondary summary. Example: 50 points for mandatory requirement satisfaction, 25 for evidence quality/freshness, 15 for cross-source consistency and 10 for unresolved risk. Apply hard gates: a failed mandatory requirement, a confirmed debarment, or an unresolved identity conflict must force Review Required or Fail regardless of the numeric score. Weight and thresholds belong to the tender rule set and must be visible to the officer.

RISK LEVEL	TRIGGER	SYSTEM ACTION
Low	All mandatory rules pass; evidence current; no material conflicts	Recommend Pass; officer confirms
Medium	Non-mandatory gap, minor staleness or low-confidence field	Recommend Clarify; show fix or confirmation
High	Mandatory gap, material mismatch, source conflict or suspicious document signal	Recommend Review Required; route to senior review
Critical	Confirmed debarment/blacklisting or identity/fraud concern supported by authoritative evidence	Block automatic recommendation; require officer/legal process

8 Security, privacy and governance

BidSure processes PAN, GST, financial, employment and identity-linked data. Security is therefore part of the product design, not a later deployment task.

CONTROL AREA	REQUIRED DESIGN	MVP EVIDENCE
Access control	Separate bidder, procurement officer, reviewer, auditor and administrator roles; least privilege; MFA/SSO seam	RBAC middleware and role test cases
Data minimization	Store only fields required for the tender rule; mask PAN/Aadhaar-like identifiers in UI; avoid sending raw documents to external models by default	Data inventory and masking utilities
Consent and authority	Capture bidder consent/authority basis, purpose, actor, timestamp and scope before sensitive lookups	Consent record linked to verification job
Encryption and secrets	TLS in transit; encrypted object storage; secrets outside code; rotate adapter credentials	Environment/secrets checklist; no keys in repository
Model governance	Prompt/version registry, model output stored as advisory evidence, confidence thresholds, human review for low confidence	Replayable extraction record and test set
Auditability	Append-only event trail with case, user, source, rule and before/after hash	Audit query and export
Retention and deletion	Policy by document type and legal purpose; deletion workflow with hold exceptions	Retention configuration and delete test
Incident handling	Source outage, credential leak, data breach, model error and false-positive escalation paths	Runbook and visible incident status

Security boundary for the demo

Use synthetic bidder data and mock/sandbox responses. Label every source in the UI. Do not scrape portals, bypass authentication or imply access to authenticated government records. The demo should prove the data model and decision controls; production access would require formal onboarding, provider approval, security review and operational ownership.

9 MVP scope and delivery plan

The best SIH MVP is narrow, end-to-end and demonstrable. It should complete one representative tender case with several positive, negative and ambiguous findings.

SPRINT	SCOPE	DONE WHEN
Day 1: foundation	React shell, Node.js/Express API, case/tender/bid schemas, object storage, role model, audit event helper	A case can be created and all artifacts have stable IDs
Day 2: intelligence	PDF/image ingestion, OCR, classification, extraction, requirement editor	Reviewer can inspect and correct extracted fields and requirements
Day 3: verification	Mock GST/Udyam/DigiLocker adapters, cross-document checks, deterministic rule engine	A sample bid produces clause-level statuses with evidence links
Day 4: decision console	Score/risk summary, finding filters, clarification workflow, officer decision and export	Officer can make and sign a decision with reason
Hardening	Failure simulations, stale source, low OCR confidence, conflicting names, duplicate document, audit replay	Demo shows safe handling of edge cases and no silent passes

Demo scenario set

- Pass case: active GST, matching Udyam, valid authorization and all mandatory declarations.
- Clarify case: OCR confidence below threshold, expired certificate and one missing exemption document.
- Conflict case: legal name differs across GST, Udyam and uploaded certificate.
- Unavailable-source case: adapter timeout returns Pending Re-verification, not Pass.
- Integrity case: duplicate certificate hash and suspicious metadata create a review finding, not an automatic fraud conclusion.

10 Evaluation, impact and limitations

Impact should be measured with a controlled benchmark rather than asserted. Compare BidSure against a fixed manual-review protocol on the same anonymized tender packets.

MEASURE	METHOD	TARGET FOR MVP VALIDATION
Extraction precision/recall	Field-level comparison against a reviewed ground-truth set	>= 95% on selected high-value fields; report by document type
Requirement extraction acceptance	Reviewer accepted vs corrected AI proposals	>= 85% accepted without semantic correction
Finding correctness	True positive, false positive and false negative counts by rule	100% of mandatory-rule outcomes reviewed; minimize false negatives
Review time	Median time from case open to officer-ready recommendation	Demonstrate material reduction on benchmark set; do not generalize from one case
Traceability completeness	Sample every finding and follow evidence links to source and rule version	100% of findings have provenance or explicit unavailable reason
Resilience	Inject source timeout, stale response, OCR failure and duplicate inputs	No silent Pass; correct status and audit event for every scenario

Known limitations

- A mock adapter demonstrates interface readiness, not production verification authority.
- OCR and LLM extraction accuracy varies by document quality, language and layout.
- A fraud signal is not a legal finding; procurement, vigilance and legal workflows remain human-led.
- Compliance rules change. A rule registry with effective dates and owner approval is mandatory for any pilot.
- The final qualification/disqualification decision remains with the Procurement Officer, as required by the problem statement.

Appendix A Risk register

RISK	LIKELIHOOD / IMPACT	MITIGATION	OWNER
Government API access not approved	High / High	Mock/sandbox adapters; formal onboarding plan; source capability registry	Platform lead
Stale or unavailable source	Medium / High	Freshness windows, explicit Pending state, retry/backoff and officer escalation	Verification lead
LLM misreads clause	Medium / High	Source-cited extraction, reviewer approval and deterministic rule execution	Tender intelligence lead
OCR misreads identifier	Medium / High	Confidence threshold, field validation and manual review queue	Document intelligence lead
False positive fraud signal	Medium / High	Use signal language, preserve evidence, require human/vigilance review	Risk lead
Sensitive-data leakage	Low / Critical	Synthetic demo data, minimization, masking, encryption, RBAC and logging	Security lead
Rule change after evaluation begins	Medium / High	Rule and tender versioning; freeze active evaluation snapshot	Product owner
Dashboard score hides hard failure	Medium / High	Hard gates and clause-level status visible above aggregate score	Compliance lead

Appendix B Research references

The following official sources informed the research sections and should be re-checked before any production integration or legal claim.

GeM General Terms and Conditions 4.0, Version 1.7

https://assets-bg.gem.gov.in/resources/upload/shared_doc/gtc/general-te-1675401798.pdf

Department of Expenditure, General Financial Rules 2017

<https://doe.gov.in/files/inline-documents/GFR2017.pdf>

API Setu overview and integration flow

<https://docs.apisetu.gov.in/document-central/explore-apisetu/Overview.html>

API Setu Terms of Use

<https://docs.apisetu.gov.in/document-central/terms-of-use/>

API Setu official SOP supplied in source set

<https://cdn.apisetu.gov.in/portal/assets/SOP-APISETU.pdf>

MeitY Digital Personal Data Protection Rules 2025

<https://www.meity.gov.in/documents/act-and-policies/digital-personal-data-protection-rules-2025-gDOxUjMtQWa?pageTitle=Digital-Personal-Data-Protection-Rules-2025>

Udyam Registration official portal

<https://udyamregistration.gov.in/>

GST official taxpayer status guidance

<https://tutorial.gst.gov.in/offlineutilities/gsterrorandresolution/gstissuesandsuggestedolutions.pdf>

PIB GeM factsheet supplied in source set

<https://pib.gov.in/Pressreleaseshare.aspx?PRID=1776721>

Appendix C Recommended next decisions

- Freeze the MVP tender scenario and rule catalogue before coding.
- Assign one owner for rule versioning and one owner for source/adapters.
- Define the exact mock response contract for GST, Udyam and DigiLocker before frontend work.
- Prepare an anonymized ground-truth document pack and edge-case scripts.
- Use the officer decision and audit export as the final demo moment.

End of report